Part 1 - AI Project Functional Methodologies

Le Linh Long, Truong Kim Tan, Hamza El Hamdi

1 Project Strategy

1.1 Strategic Objectives

In this project, we set the strategic objectives are:

- Identify customers at high risk of churning.
- Reduce churn rates by enabling targeted retention strategies.
- Enhance the overall user experience and customer lifetime value.

1.2 Key Performance Indicators

We will divide our project success metrics into two main categories: Business KPIs and Model Performance Metrics.

For the business KPIs, we set the following metrics followed by Content Square [1]:

- **Customer Churn Rate:** Measures the percentage of users who stop using the platform. We use this as our main baseline to see if the AI system helps reduce user loss over time.
- **Customer Lifetime Value (LTV):** Estimates how much total money a single user brings in over time. This helps us understand if our actions help users stay longer and increase revenue.
- **Customer Satisfaction Score (CSAT):** Measures how happy users are after getting help from customer support. This shows whether our retention efforts improve how users feel about the service.

For model performance, we utilize core Machine Learning metrics to evaluate the performance of our algorithm:

- **Recall:** Measures how many of the actual customers who left were correctly found by the model. This is important because missing a customer who is about to leave is more costly than contacting someone who would have stayed.
- **Precision:** Measures how many of the customers predicted to leave actually ended up leaving. High precision means we avoid spending money on customers who were not going to leave anyway.
- **F1-Score:** Combines Precision and Recall into one score. It helps us get a balanced view of the model's performance.

1.3 How Can AI Help Improve Customer Retention?

AI can improve customer retention by analyzing large amounts of user data such as interactions, purchase history, and other signals. It can detect early signs that a customer may be unhappy or about to leave.

Unlike rule-based systems, AI can learn and adapt as customer behavior changes. This allows it to make real-time and personalized actions that solve customer problems early, before they decide to leave.

2 Project Design

2.1 Data

This project uses both internal and external data sources. Internal data includes user interactions such as product views and searches, purchase history, customer demographics, seller activities, order processing records, customer support tickets, and customer reviews. External data includes market trends, sentiment data from third-party review platforms, competitor pricing information, economic and demographic statistics, social media data, and weather data.

Several challenges may arise when working with these data sources. One challenge is data integration, as information may be stored across different systems such as SQL databases, NoSQL databases, cloud storage, and data lakes. Another challenge is data quality, since the data may contain missing values, inconsistencies, or outliers that can affect model performance.

Data privacy and security are also important considerations. Customer information must be handled carefully and comply with regulations such as GDPR and CCPA, especially when combining internal data with data from external sources. In addition, the system should support real-time processing so that customer behavior can be analyzed quickly and efficiently while maintaining scalability as the platform grows.

2.2 Models

The choice of model depends on the size of the dataset. If the dataset is relatively small, Machine Learning models such as LightGBM will be used because they provide strong performance and support GPU acceleration for faster training. If the dataset is very large and contains complex patterns, Deep Learning models will be considered instead.

During model training, we will carefully structure our training pipeline to prevent data leakage, ensuring test set information remains isolated and results reflect true model performance.

For model validation, the dataset will be split into separate training and validation sets. Validation ML metrics mentioned will be used to evaluate the model.

For model testing, both prediction performance and inference time will be evaluated. This helps ensure that the model is accurate while also being fast enough for real-world deployment. MLflow will be used for model versioning and experiment tracking. This allows different model versions, parameters, and evaluation results to be recorded and compared easily.

For model serving, Flask and Docker will be used to deploy the model as a scalable API service. Flask will handle prediction requests, while Docker will provide a consistent and portable deployment environment.

2.3 Deployment

2.3.1 Deployment Strategies

The model will be deployed gradually to reduce risk and ensure stable performance in the production environment.

The first deployment stage will use Shadow Mode. In this approach, the new model runs alongside the existing system and generates predictions without affecting customer-facing decisions. This allows the team to evaluate the model's performance on real-world data and identify potential issues before deployment.

After successful testing in Shadow Mode, the model will be deployed using A/B Testing. A small group of users will receive AI-driven retention actions, while another group will continue using the existing system. The results from both groups will be compared using business metrics such as customer retention rate, churn rate, and customer lifetime value. If the new model shows positive results, it can be rolled out to all users.

Initially, the system may use batch predictions, where churn risk scores are generated on a scheduled basis, such as once every night. This approach is simpler to implement and requires fewer computing resources. As the platform grows and requires faster responses, the system can be upgraded to real-time inference, allowing retention actions to be triggered immediately when a customer shows signs of leaving.

2.3.2 Considerations for the Production Environment

Several factors must be considered when deploying the model in a production environment.

Scalability is important to ensure the system can handle increasing numbers of users and prediction requests. Technologies such as Docker containers and cloud-based infrastructure can help support future growth and automatically allocate resources when traffic increases.

Latency is another important factor. Prediction requests should be processed quickly so that users do not experience delays when interacting with the platform. Low latency is especially important if real-time retention actions are implemented.

Integration with existing business systems is also required. The prediction results should be connected to marketing automation tools, customer relationship management systems, and customer support dashboards. This allows business teams to use the model's predictions effectively when creating retention campaigns and supporting customers.

Finally, the deployed model should be continuously monitored to track prediction accuracy, system performance, and business impact. Monitoring helps detect performance degradation and ensures that the model continues to provide value over time.

2.4 Monitoring

2.4.1 Monitoring Model Performance

After deployment, the model's performance must be continuously monitored to ensure that it remains accurate and useful.

Technical metrics such as Precision, Recall, and F1-Score will be tracked by comparing the model's predictions with actual customer outcomes. These metrics help evaluate how well the model identifies customers who are likely to churn.

Business metrics will also be monitored to measure the real impact of the system. Important metrics include customer churn rate, customer lifetime value, customer satisfaction score, and the return on investment (ROI) of retention campaigns. Dashboards and visualization tools can be used to help business teams track these metrics over time.

In addition, the quality of incoming data will be monitored to detect missing values, unusual patterns, or changes in data distributions that may affect model performance.

2.4.2 Handling Model Drift and Maintaining Accuracy

Customer behavior can change over time, which may reduce the accuracy of the model. Therefore, mechanisms should be in place to detect and handle model drift.

Data drift occurs when the distribution of input features changes compared to the training data. Concept drift occurs when the relationship between customer behavior and churn changes. Automated monitoring tools can be used to detect these changes and generate alerts when significant drift is observed.

To maintain performance, the model will be retrained periodically using the latest available data. Retraining may be scheduled on a regular basis, such as monthly, or triggered automatically when drift exceeds a predefined threshold.

Feedback from customer support teams and marketing teams will also be incorporated into the monitoring process. Their observations can help identify new customer behaviors and reasons for churn that may not be captured by the current model. This human feedback can be used to improve future versions of the model and keep it aligned with business needs.

3 Project Team

3.1 Roles and Expertise Required

The Data Engineer will be responsible for building data pipelines, maintaining data quality, managing data storage systems, and deploying machine learning models. This role will also handle monitoring, automation, and the CI/CD process to ensure smooth operation in production.

The ML Engineer will focus on data analysis, feature engineering, model development, training, and evaluation. This role is responsible for selecting the most suitable machine learning models and improving their performance.

The Product Owner and Business Analyst will ensure that the project aligns with business objectives. This role will define KPIs, gather requirements from stakeholders, and work closely with marketing and customer support teams to translate model predictions into effective retention strategies.

3.2 Ensuring Cross-Functional Collaboration

Successful project delivery requires strong collaboration between technical and business teams. Regular meetings such as daily stand-ups, sprint planning sessions, and sprint reviews will be conducted to keep all team members aligned.

Shared dashboards will be created so that business stakeholders can monitor business outcomes while technical teams can track model performance. Project documentation will be stored in a centralized platform to ensure that all team members have access to the latest information.

3.3 Skills and Expertise Needed

The project requires technical skills in Python, SQL, machine learning frameworks, cloud platforms, data engineering, and model deployment technologies. Knowledge of tools for data processing and workflow automation is also important.

From a business perspective, team members should understand e-commerce metrics, customer retention strategies, marketing campaigns, communication, and project management practices.

3.4 Ensuring Alignment with Project Strategy

To ensure that the project remains focused on business goals, clear objectives and key results will be defined at the beginning of the project. These objectives will be directly linked to reducing customer churn and improving customer retention.

Regular sprint reviews will be conducted to demonstrate project progress and evaluate the impact of the model on business metrics. This helps ensure that technical development continues to support business needs.

3.5 Collaboration with Other Departments

Collaboration with other departments is essential for the success of the project.

The marketing team will use churn predictions to identify customers who may leave the platform and create targeted retention campaigns. The data science team will help explain the factors that contribute to churn so that marketing strategies can be more effective.

The customer support team will use churn risk scores and recommendations to provide better support to at-risk customers. This allows support agents to take proactive actions that may improve customer satisfaction and retention.

4 Project Governance Communication

4.1 Key Stakeholders and Communication Plan

Effective communication is important to ensure that all stakeholders remain informed and aligned with project goals.

Executive leadership will act as the project sponsor and receive monthly updates on project progress, business impact, return on investment (ROI), and alignment with organizational objectives.

Product managers and marketing executives will participate in bi-weekly meetings to discuss model insights, customer retention performance, campaign results, and new business requirements. These meetings help ensure that the AI solution continues to address business needs.

Technology and data teams will hold weekly technical meetings to discuss data pipelines, infrastructure, model improvements, deployment activities, and any technical challenges that need to be addressed.

4.2 Governance Structure

A steering committee will be established to oversee the project at a strategic level. The committee will meet monthly or quarterly to review project milestones, approve major decisions, allocate resources, and resolve significant issues that may affect project success.

A data governance board will be responsible for ensuring compliance with data privacy regulations and maintaining data quality standards. This group will oversee how customer data is collected, stored, processed, and used throughout the project lifecycle.

4.3 Communicating Model Outputs

Different stakeholders require different types of information from the AI system.

Technical teams will receive detailed reports on model performance, feature importance, deployment status, and system monitoring. These reports will help data scientists and engineers evaluate and improve the model over time.

Non-technical teams will be provided with simple and easy-to-understand dashboards that focus on business outcomes. Examples include customer churn trends, the main factors contributing to churn, retention campaign performance, and estimated revenue saved through retention efforts.

By presenting information in a clear and understandable format, all stakeholders can make informed decisions without requiring advanced knowledge of machine learning concepts.

5 AI Project Management Methodology

5.1 Chosen Methodology: Agile (Scrum with Data Science Adaptations)

This project will use an Agile approach, specifically Scrum adapted for data science and machine learning work.

Agile is well suited for AI projects because there is often uncertainty in the data and model performance. Machine learning development is highly experimental and requires frequent testing and adjustment. Unlike traditional software projects, models improve through iteration rather than a fixed development path.

Using Agile, the team can quickly build a basic working model (minimum viable product), gather feedback from business stakeholders such as marketing and product teams, and then improve the model over time. This approach also allows the team to adapt when new data becomes available or when business requirements change.

5.2 Potential Risks and Mitigation Strategies

Several risks may affect the success of the project.

One risk is poor data quality or lack of sufficient data. This can be reduced by performing a detailed data audit early in the project and implementing strong validation checks in the data pipelines.

Another risk is model underperformance, such as low accuracy or weak recall. This can be addressed by starting with simple baseline models, then gradually improving performance through feature engineering and testing different algorithms.

A further risk is low user adoption, where business teams may not use the model's predictions. To reduce this risk, marketing and product teams will be involved from the beginning. In addition, model explanations such as SHAP values can be used to show why a customer is predicted to churn, making the results more transparent and easier to trust.

5.3 Handling Costs and Planning Execution

To manage resources effectively, timeboxing will be used. This means setting fixed time periods for research and experimentation to avoid spending too long on unproductive exploration.

Cloud infrastructure costs will be monitored using budget tracking tools and alerts. This helps ensure that expensive training jobs and computations remain within budget.

Finally, task prioritization will be managed through a structured backlog. Each task will be evaluated based on its expected business value, ensuring that the team focuses on work that delivers the most impact first.

References

[1] Contentsquare Content Team. 7 Metrics and KPIs to Calculate and Track Customer Retention, 2024. `https://contentsquare.com/guides/customer-retention/metrics/`.